

Simon Lin

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EDUCATION

Toronto Metropolitan University (Ryerson University)

Bachelor of Science - Computer Science with Co-op

Toronto, ON

Sep 2021 – Present

- GPA: 4.26/4.33, Dean's List (2021-2025)
- Relevant Courses: Software Engineering, Data Structures, Artificial Intelligence, Web Development, OS.

EXPERIENCE

Software Engineer Co-op

Magna International

May 2025 – Dec 2025

Newmarket, ON

- Developed and deployed full-stack web application (**Python + Flask + React.js**) using **AWS** to accelerate Finite Element Analysis (FEA) processes to support 15+ global engineers.
- Implemented a 3D FEA model viewer using **Three.js**, to visualize complex tensile stresses in a web browser.
- Built a CATIA automation tool, reducing manual data entry time by 70% in CAD files and improving efficiency for cross-functional engineering teams.

Software Engineer Co-op

SOTI

Apr 2024 – Aug 2024

Mississauga, ON

- Developed new web application features in **C#** and **.NET**, adhering to technical and customer specifications.
- Implemented **REST** APIs to improve response times between client applications and backend services by **35%**.
- Designed and implemented relational database schemas using **Microsoft SQL Server**, normalizing data models to reduce redundancy and improved query performance across 5+ core product tables.
- Collaborated in an Agile CI/CD pipeline, participating in weekly planning and code reviews to ship features.

Software Automation Engineer Co-op

SOTI

Sep 2023 – Apr 2024

Mississauga, ON

- Created automated test plans, test cases and unit tests in **C#**, **Selenium** and **Microsoft Azure DevOps**.
- Discovered and resolved over **155** defects and performance issues across product UI and dashboards.
- Lowered SQL database retrieval time by **27%** by optimizing legacy queries to clean, transform & aggregate data.

Python Coding Instructor

The STEAM Project

Jul 2021 – Aug 2021

Richmond Hill, ON

- Taught 20 summer camp students from grades 7-8 a creative curriculum centered on Python fundamentals.
- Collaborated with co-instructors to create an educational and interactive coding curriculum on a weekly basis.

EXTRACURRICULARS

TMU Formula SAE

Embedded Firmware Engineer

Sep 2021 – Present

Toronto, ON

- Developed firmware in C for Arduino microcontrollers to control and poll vehicle subsystems.
- Developed a telemetry dashboard with Python libraries to support testing workflows and enable real time diagnostics.

PROJECTS

NBAction | [GitHub Repository](#)

Sep 2023 – Dec 2024

- Developed and trained a deep-learning model capable of detecting basketball actions to analyze live basketball games.
- Optimized a video processing pipeline using dynamic frame sampling algorithms to reduce live video latency by 68%.
- Performed model fine-tuning to achieve an overall detection rate of 92% across all object classes.

SKILLS

Programming Languages: Python, JavaScript, C, C#, TypeScript, Java, SQL, Bash

Libraries & Frameworks: React, Vue.js, Three.js, Flask, OpenCV, .NET, Selenium, Postman

Developer Tools: AWS, Azure Cloud, Git, Jira, Redis, Docker, Azure DevOps, GitHub